

Monday Warm Up: Ohm's Law, Energy and Power in DC Circuits.

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1 Memory Bank

- $I = \Delta Q / \Delta t$... Definition of current.
- $\Delta V = IR$... Ohm's Law.
- $R = \rho(L/A)$... Resistivity, ρ , and resistance, R , for a conductor with cross-sectional area A and length L .
- $\rho = \rho_0(1 + \alpha\Delta T)$... Resistivity depends on changes in temperature.
- $R = R_0(1 + \alpha\Delta T)$... Resistance depends on changes in temperature.
- $P = \Delta E / \Delta t$... Power is energy or work per unit time.
- $P = I^2 R = V^2 / R = IV$... Formulae for power consumption in a DC circuit.

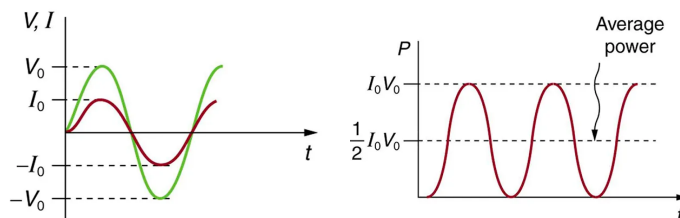


Figure 1: (Left) Voltage and current vs. time in an AC circuit. (Right) Average power in AC circuit.

2 Ohm's Law

1. What current flows through the bulb of a 3.00-V flashlight when its hot resistance is $3.60 \, \Omega$?
2. What is the effective resistance of a car's starter motor when 150 A flows through it as the car battery applies 11.0 V to the motor?

2. (a) Verify that the units of a volt-ampere are watts, as implied by the equation $P = IV$. (b) Show that the units $V^2/\Omega = 1 \, \text{W}$, as implied by the equation $P = V^2/R$. (c) Show that the units $A^2\Omega = 1 \, \text{W}$, as implied by the equation $P = I^2 R$.

3 Electric Power and Energy

1. A charge of 4.00 C passes through a pocket calculator's solar cells in 4.00 h. What is the power output, given the calculator's voltage output is 3.00 V?

3. Verify the energy unit equivalence that $1 \, \text{kW hr} = 3.60 \times 10^6 \, \text{J}$.
4. An electric water heater consumes 5.00 kW for 2.00 h per day. What is the cost of running it for one year if electricity costs 12.0 cents per kilowatt hour?

5. Consider Fig. 1. (a) Let $V_0 = 120$ Volts, and $I_0 = 50$ mA. What is the average power? (b) Let $T = 16.7$ ms be the *period* of the wave, the time between peaks in voltage. What is $f = 1/T$, the *frequency* of the wave?